

ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	35.0	Gender	Male
Height (cm)	162.0	Weight (kg)	85.0
BMI	32.39	Product Info 2	987 65 4321A
Occupation	prof-specialty	Annual Income	Rs. 50,000.00
Insurance Type	Health	Coverage Amount	Rs. 650,000.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 3 of 8
Risk Category	Medium Risk
Recommended Annual Premium	Rs. 14,000.00

3. AI Actuarial Analysis

Actuarial Report: Insurance Application Evaluation

Applicant: officer@shield.com

Product: Health Insurance, Coverage: Rs. 6,50,000

ML Predicted Risk Class: 3 (out of 8), Confidence: 55.5%

Calculated Annual Premium: Rs. 14,000.00

1. Risk Assessment

The applicant presents a mixed risk profile, with several positive indicators counterbalanced by a significant health concern.

* **Age (35):** Moderate risk. Within a standard working age bracket, generally associated with lower acute health risks compared to older age groups.

* **BMI (32.39):** **High Risk.** This BMI falls into the Obese (Class I) category. Obesity is a significant health risk factor, increasing the likelihood of conditions such as diabetes, cardiovascular disease, hypertension, and certain musculoskeletal issues, which are highly relevant for a Health insurance policy.

* **Smoker (0):** Low Risk. Non-smoker status is a strong positive health indicator.

* **Occupation (prof-specialty):** Generally Low to Moderate Risk. This occupation category typically involves professional roles, often desk-bound, with lower physical hazards compared to manual labor.

* **Income (Rs. 50,000):** Moderate. Assuming this is monthly income (Rs. 6,00,000 annually), the requested coverage of Rs. 6,50,000 is reasonable relative to income for health insurance.

* **Previous Claims (0) & Family History (0):** Low Risk. Absence of past claims and adverse family history are very positive indicators.

* **Exercise (1) & Alcohol (0):** Low Risk. Stated regular exercise and no alcohol consumption are positive lifestyle choices, though the high BMI might suggest dietary factors or other underlying issues.

* **Gender (Male):** Standard demographic factor, not a primary risk driver in isolation for this age.

Overall Risk Profile: The applicant exhibits a moderate-to-elevated risk profile primarily due to their high BMI, despite otherwise favorable lifestyle factors and claims history. The obesity is the most critical risk factor for a Health insurance product.

2. Comparative Analysis

The current application (Risk Class 3, Premium Rs. 14,000) shows both similarities and differences with historical approved cases:

* **Compared to POL-2626 (Risk Class 2, Premium Rs. 12,000, Similarity: 42.8%):**

* The current applicant is predicted at a slightly higher risk class (3 vs 2) with a higher premium (Rs. 14,000 vs Rs. 12,000). This difference is likely attributable to the applicant's significantly higher BMI (32.39). While other factors are positive, the obesity likely elevates the overall risk compared to a lower-risk client profile. The 42.8% similarity indicates significant differentiating factors.

* **Compared to POL-25755 (Risk Class 5, Premium Rs. 20,000, Similarity: 40.0%):**

* The current applicant is predicted at a significantly lower risk class (3 vs 5) with a considerably lower premium

(Rs. 14,000 vs Rs. 20,000). This aligns with the applicant's positive factors such as non-smoker status, no previous claims, no adverse family history, and positive lifestyle indicators, which would typically lead to a more favorable risk assessment than a Risk Class 5 individual.

* **Compared to POL-31852 (Risk Class 7, Premium Rs. 30,000, Similarity: 38.8%):**

* The current applicant is at a much lower risk class (3 vs 7) and premium (Rs. 14,000 vs Rs. 30,000). The significant difference in risk and premium is expected, as Risk Class 7 represents a considerably higher risk profile, which the current applicant's overall positive lifestyle and claims history do not reflect.

3. Anomaly & Fraud Detection

* **Significant High BMI:** The applicant's BMI of 32.39 is classified as Obese (Class I), which is a clear and significant health anomaly. While the applicant reports exercising and no alcohol, the high BMI indicates either dietary issues, insufficient exercise to manage weight, or other metabolic factors. This is a primary concern for health insurance.

* **Low ML Prediction Confidence (55.5%):** The relatively low confidence level of the Machine Learning classifier suggests that the applicant's profile may not perfectly align with typical cases within Risk Class 3, or it presents borderline characteristics. This lack of strong confidence from the automated model indicates potential for misclassification and warrants further scrutiny.

* **Coverage vs. Income (Assumed Monthly Income):** Assuming the Rs. 50,000 income is monthly (i.e., Rs. 6,00,000 annually), the coverage amount of Rs. 6,50,000 is approximately 1.08 times the annual income. This is a reasonable and not unusual coverage level for a health insurance policy and does not flag as a financial anomaly. If the income were annual, it would be a significant anomaly, but this is less likely for a "prof-specialty" occupation.

* **No other immediate red flags:** No unusual claims history, inconsistent smoking/alcohol status, or suspicious family history were detected.

4. Premium Recommendation

The calculated annual premium of Rs. 14,000.00 appears numerically consistent with the predicted Risk Class 3 when viewed in comparison to the provided historical data.

* It is higher than the Risk Class 2 case (Rs. 12,000), which is justifiable given the higher BMI.

* It is lower than the Risk Class 5 and 7 cases (Rs. 20,000 and Rs. 30,000 respectively), which is consistent with the applicant's better overall risk profile in many aspects.

However, the **low prediction probability (55.5%)** from the ML model is a critical factor. While the Rs. 14,000 premium is derived from a Risk Class 3 assignment, the model's low confidence indicates uncertainty in this classification. This suggests that while the premium aligns with the *assigned* risk class, the *assignment itself* might be tenuous due to the unique combination of the applicant's attributes (e.g., healthy lifestyle reports despite high BMI). This uncertainty necessitates caution regarding the final premium acceptance without further validation.

5. Underwriting Decision Recommendation

****Recommendation: Refer for Manual Review****

****Justification:****

Despite several positive attributes (non-smoker, no claims, no family history, reported healthy lifestyle), the

applicant presents a significant health risk due to their ****Obese (Class I) BMI of 32.39****. This factor alone warrants careful consideration for a Health insurance policy.

Furthermore, the ****Machine Learning classifier's low confidence level of 55.5%**** for Risk Class 3 indicates that the automated system is not highly certain about its assessment. This uncertainty, combined with the primary health anomaly (high BMI), suggests that the application falls into a complex category that requires human expertise.

A manual review would allow an experienced underwriter to:

1. ****Investigate the high BMI:**** Understand the underlying causes of obesity despite reported exercise and no alcohol. This might involve requesting further medical information, such as recent health check-ups, blood reports, or a doctor's statement regarding their weight management efforts and overall health status.
2. ****Validate lifestyle claims:**** Assess the consistency of reported exercise and alcohol habits with the high BMI.
3. ****Confirm appropriate risk class:**** Determine if Risk Class 3 is truly appropriate, or if a higher risk class (with a corresponding premium adjustment) or specific exclusions/loading might be necessary given the obesity.

Direct approval is not recommended due to the unaddressed health risk and ML uncertainty. Outright rejection is premature given the positive factors that may mitigate some of the risk. A manual review ensures a comprehensive and accurate risk assessment.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **Rs. 14,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.